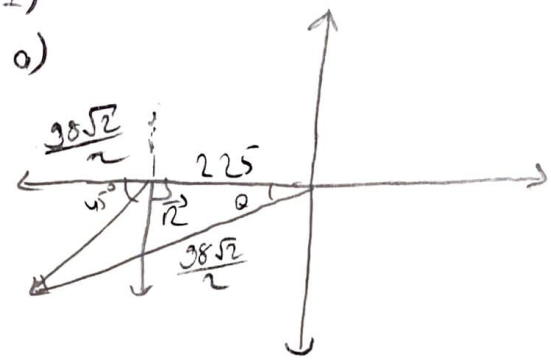
G131210303
Boris Yılmaz
1/A

B. Yılmaz

1)

a)



$225 + \frac{98\sqrt{2}}{2} = 294.3\,km$ batı

$\frac{98\sqrt{2}}{2} = 69.3\,km$ güney

$R = \sqrt{(294.3)^2 + (69.3)^2} = 302\,km$

$\theta = \tan^{-1}\left(\frac{69.3}{294.3}\right) = 13°$ → güneybatı

b) $\vec{A}.\vec{B} = (3\hat{i} - 2\hat{j}).(2\hat{i} + a\hat{j}) = 6 - 2a = 0 \Rightarrow a = 3$

c) $2\vec{A} - \vec{B} = 2(3\hat{i} - 2\hat{j}) - (2\hat{i} + 3\hat{j}) = 6\hat{i} - 4\hat{j} - 2\hat{i} - 3\hat{j} = 4\hat{i} - 7\hat{j}$

$3\vec{A}.2\vec{B} = 6.(\vec{A}.\vec{B}) \Rightarrow \vec{A}.\vec{B} = 0$ olduğundan $3\vec{A}.2\vec{B} = 0$

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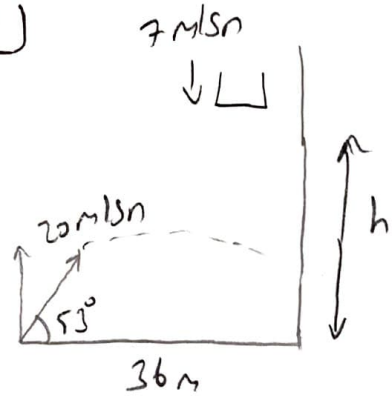
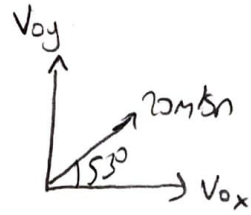
2)





$V_{0x} = 20 . \cos 53 = 12,02$ "

$V_{0y} = 20 . \sin 53 = 15,96$ "

$x = V_{0x}\, t \Rightarrow 36m = 12,02 . t \Rightarrow t = 2,9950$

$y = V_{0y} . t - \frac{1}{2} g t^2 = (15,96 . 2,9950) - \frac{1}{2} . 9,8 . (2,9950)^2$

$y = 3,8472$ noktasından

$7 . (2,9950) = 20,965$ m yol alır.

$h = 20,965 + 3,8472 = 24,8122$ yüksekliğinden geçer.

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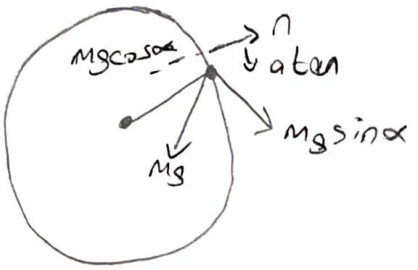
3)



$\sum F_y = m a_y$

$mg\cos\alpha - n = m \frac{v_2^2}{R}$

$n = 0 \Rightarrow mg\cos\alpha = m \frac{v_2^2}{R}$

$y_1 = R \quad y_2 = R.\cos\alpha$

$v_2^2 = R.g.\cos\alpha$

$a_{rad} = \frac{v^2}{R}$

$K_1 + U_1 = K_2 + U_2 \qquad K_1 = 0 \quad K_2 = \frac{1}{2} m v_2^2$

$U_1 = mg\,y_1 = mgR \qquad U_2 = mg\,y_2 = mgR\cos\alpha$

$mgR = \frac{1}{2} m v_2^2 + mgR\cos\alpha$

$v_2^2 = 2gR(1-\cos\alpha)$

$Rg\cos\alpha = 2gR(1-\cos\alpha)$

$\cos\alpha = 2 - 2\cos\alpha$

$3\cos\alpha = 2$

$\cos\alpha = 2/3$

$\alpha = 48{,}2^\circ$ //

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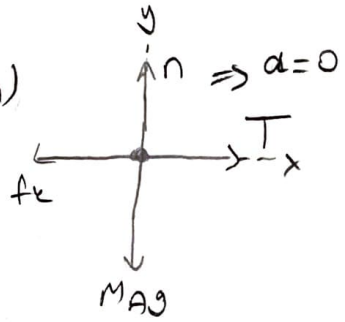
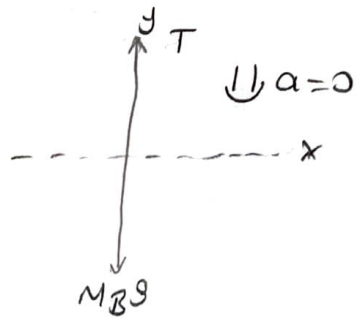
4) $M_A = \frac{W_A}{g} = \frac{45}{9,8} = 4,59\,kg = M_{kedi}$

$M_B = \frac{W_B}{g} = \frac{25}{9,8} = 2,55\,kg$

a)





b) <u>A blosu için</u> <u>B blosu için</u> ⇒ hız sabitse ivme 0

$\Sigma F_x = M_A a_x$
$\Sigma F_y = M_A a_y$

$T - F_k = M_A \cancel{a}^{0}$

$\Sigma F_y = M_B a_y$
$M_B g - T = M_B \cancel{a}^{0}$

$T = M_B g = W_B = 25N$

$f_k = T = 25N$

$\Sigma F_y = 0$
$n - M_A g = 0$
$n = M_A g$
$f_k = n\mu_k$
$f_k = M_A g \mu_k = 25N$

$\mu_k = \frac{25N}{M_A g} = \frac{25N}{45N} = 0,556$

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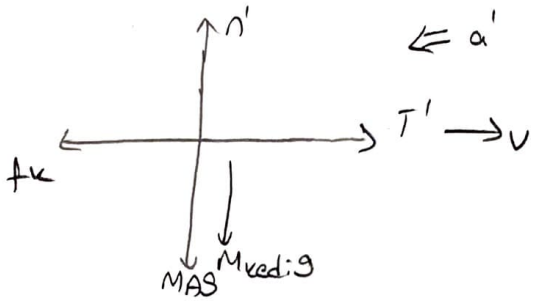
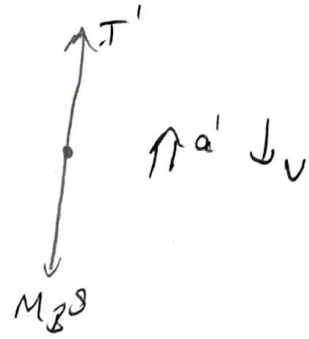
c) Kedi uyuduktan sonra





$f_k - T' = (M_A + M_{kedi})a'$

$T' - M_B g = M_B a'$

$f_k = n'\mu_k = (M_A g + M_{kedi} g)\mu_k$
$= (45+45)\cdot\frac{5}{9}$

$= 50N$

$(T' \neq T)$

$50 - T' = (M_A + M_{kedi})a'$
$50 - (M_B a' + 25) = (M_A + M_{kedi})a'$

$25 = (M_A + M_B + M_{kedi})a' \Rightarrow a' = \frac{25}{4,5g + 2,55 + 4,5g} = \frac{25N}{11,73 kg}$

$a = 2,13 \ m/s^2$

B $-\hat{j}$ yönünde